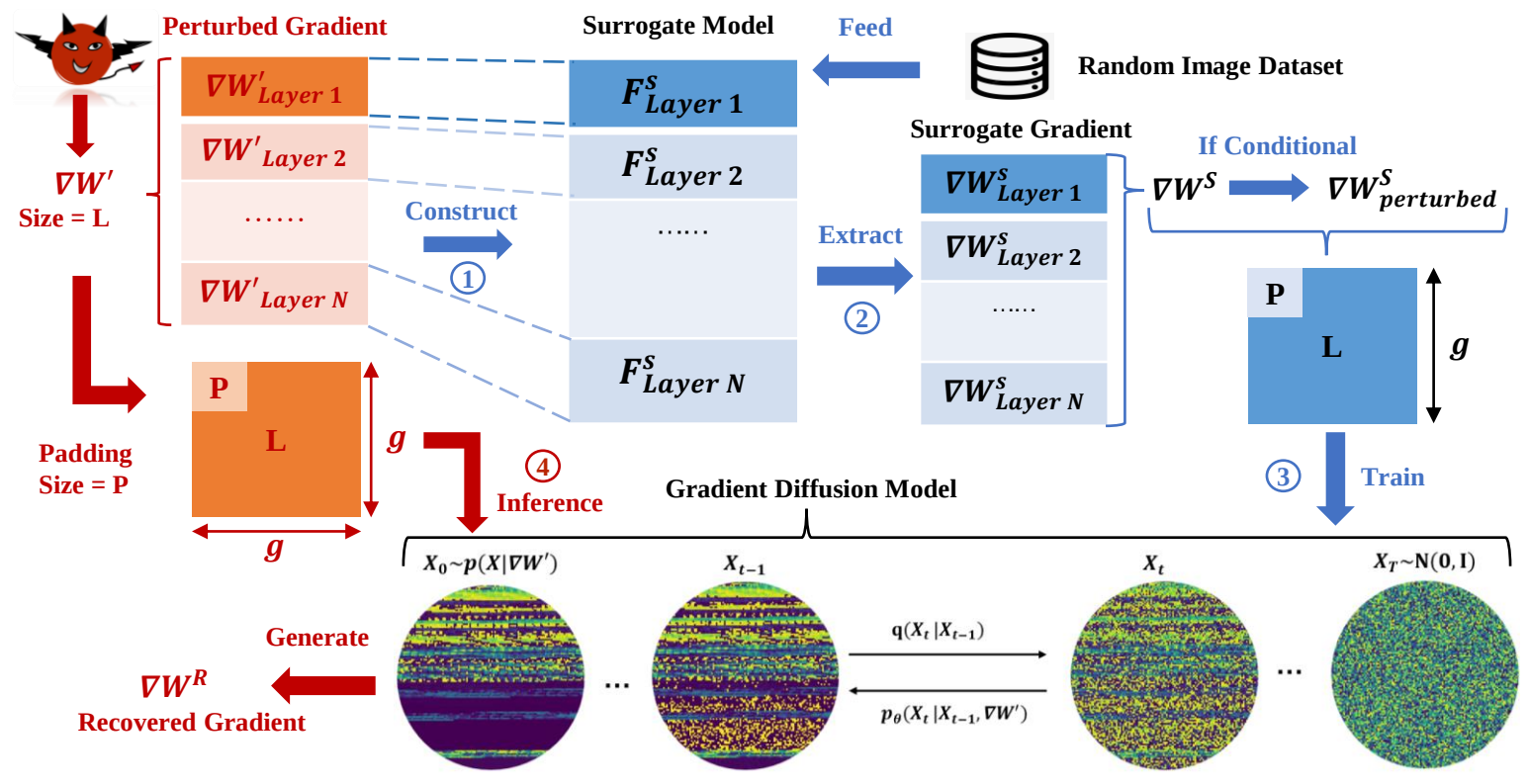

Perturbed Gradient
Surrogate Model
Feed
Random Image Dataset
$\nabla W'_{Layer\ 1}$
$\nabla W'_{Layer\ 2}$
.....
$\nabla W'_{Layer\ N}$
$F^S_{Layer\ 1}$
$F^S_{Layer\ 2}$
.....
$F^S_{Layer\ N}$
Surrogate Gradient
If Conditional
$\nabla W^S_{Layer\ 1}$
$\nabla W^S_{Layer\ 2}$
.....
$\nabla W^S_{Layer\ N}$
∇W^S
$\nabla W^S_{perturbed}$
$\nabla W'$
Size = L
Construct
①
Extract
②
P
L
g
Padding
Size = P
④
Inference
③
Train
Gradient Diffusion Model
$X_0 \sim p(X|\nabla W')$
X_{t-1}
X_t
$X_T \sim \mathbf{N}(0, \mathbf{I})$
$q(X_t|X_{t-1})$
$p_\theta(X_t|X_{t-1}, \nabla W')$
Generate
∇W^R
Recovered Gradient